

Comprehensive

Big Data Analytics

Exam-Ready Study Notes

9 Key Topics	4 Vs of Big Data	Analytics Types	Tech Ecosystem
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"Big data" refers to high-volume, high-velocity, and high-variety information assets that demand cost-effective, innovative forms of information processing. It describes collections of datasets so large and complex that traditional data processing applications and management tools are inadequate to store or process them efficiently.

Historical Evolution of this

- **Origins:** First data centres and relational databases emerged in the 1960s–70s.
- Post-2005 explosion driven by social media, the Internet of Things (IoT), and machine learning.

Data Size Units

Unit	Abbreviation	Context
Megabyte	MB	Small files, documents
Gigabyte	GB	HD videos, software
Terabyte	TB	Enterprise databases
Petabyte	PB	Large-scale data warehouses
Exabyte	EB	Global internet traffic
Zettabyte	ZB	Total data created worldwide per year
Yottabyte	YB	Theoretical future scale

2. Characteristics of Big Data (The 4 Vs)

Volume	Massive size of data generated daily from various sources.
Velocity	Speed at which data is created and flows in real-time (logs, social media, IoT).
Variety	Heterogeneous formats — structured, unstructured, and semi-structured.
Veracity	Quality and accuracy of captured data; affects analysis reliability.

3. Classification and Types of Data

Type	Description	Examples
Structured	Fixed format; rows and columns.	RDBMS, employee tables
Unstructured	No specific form; difficult to process.	Emails, PDFs, photos, videos, web logs
Semi-structured	Mix of structured and unstructured; uses tags or markers.	XML, JSON files
Multi-structured	Multiple combined formats; non-transactional systems.	Social-media feeds, clickstreams

4. Traditional Systems vs. Big Data

Business Intelligence (BI) vs. Big Data Analytics

Feature	Business Intelligence (BI)	Big Data Analytics
Data Storage	Central server / Data Warehouse	Distributed file systems (HDFS)
Processing	Data brought to functions (offline)	Functions carried to data (Parallel Mass Processing)
Data Format	Relational, structured data	Structured, semi-structured, and unstructured
Timing	Historical data only	Historical or real-time sources

RDBMS vs. NoSQL

Feature	RDBMS	NoSQL
Data model	Structured, relational tables	Flexible (document, key-value, graph, column)
Query language	SQL	Not Only SQL — varies by database
ACID rules	Strict ACID compliance	May relax ACID for performance (BASE model)
JOINS	Supported	Not used; data is denormalised
Fault tolerance	Limited replication	Built-in replication across nodes
Best for	Structured, transactional data	Semi-structured / unstructured, large-scale data

5. Big Data Computing Models & Architecture

Computing Models

- **Scalability:** Vertical (scale up — add resources to one system) vs. Horizontal (scale out — add more nodes).
- **Cloud Computing:** On-demand shared resources over the internet. Service models: IaaS, PaaS, SaaS.
- **Grid Computing:** Distributed, geographically dispersed, heterogeneous computers on a common task.
- **Cluster Computing:** Connected nodes shifting processes for load balancing.
- **Volunteer Computing:** Resources donated by organisations or individuals for distributed projects.

5-Layer Data Architecture

L1	Sources	Identification of data sources, formats, push/pull mechanisms.
L2	Ingestion / ETL	Extract, Transform, Load — batch or real-time pipelines.
L3	Storage	Distributed file systems (HDFS) or NoSQL databases.
L4	Processing	MapReduce, Spark, or Hive — synchronous or asynchronous.
L5	Consumption	BI, analytics, visualisation, and reporting dashboards.

6. Data Quality and Pre-processing

Data must be prepared before analysis to prevent misleading decisions.

Data Quality — The 5 Rs

- **Relevancy:** Data is pertinent to the problem at hand.
- **Recency:** Data is up-to-date and timely.
- **Range:** Data covers the required scope and time period.
- **Robustness:** Data is consistent and repeatable.
- **Reliability:** Data comes from a trustworthy, accurate source.

Factors Affecting Data Quality

- **Data Noise:** Meaningless or irrelevant information contaminating the dataset.
- **Outliers:** Values that fall far outside expected ranges.
- **Missing Values:** Incomplete records that can skew analysis.
- **Duplicate Values:** Repeated entries inflating results.

Pre-processing Steps

- **Data Cleaning:** Removing or correcting incomplete, incorrect, or irrelevant data.
- **Data Enrichment / Editing:** Refining and adjusting raw datasets to improve quality.
- **Data Reduction & Wrangling:** Transforming and mapping data into simplified, ordered formats suitable for analytics.

7. Big Data Analytics

Analytics uncovers trends, patterns, and correlations to enable data-informed decisions. The general process: Collect → Process (Batch or Stream) → Clean → Analyse.

Classification of Analytics

1. Descriptive	<i>What happened?</i>	Summarises historical data to identify patterns and report on past performance.
2. Diagnostic	<i>Why did it happen?</i>	Analyses root causes using drill-down techniques.
3. Predictive	<i>What will happen?</i>	Uses ML, AI, and statistical models on historical data to forecast future behaviours.
4. Prescriptive	<i>What should we do?</i>	Goes further than predictive — generates automated decisions and recommendations.

8. Big Data Technologies & Ecosystem

Technology	Type	Key Points
Hadoop	Open-source framework	Includes HDFS (distributed storage) and MapReduce (parallel processing across clusters).
Apache Spark	Analytics engine	In-memory processing; up to 100x faster than Hadoop for certain tasks. Supports batch and stream.
Python	Programming language	Excellent for unstructured data, ML libraries (pandas, scikit-learn, TensorFlow).
R	Programming language	Geared toward statistical computing and handling structured data.
MongoDB	NoSQL database	Document-oriented, written in C++ and JavaScript. Free and highly reliable.

9. Real-World Applications & Challenges

Business Benefits

- Better customer insights and personalised marketing.
- Improved supply-chain agility and anomaly detection.
- Optimised product portfolios.
- Increased market intelligence and competitive advantage.

Industry Use Cases

Industry	Big Data Applications
Retail	Product portfolio optimisation, supply-chain monitoring, quality-deficit detection, seasonal-sales prediction.
Social Media / Marketing	Sentiment analysis, campaign ROI tracking, customer-behaviour prediction, hyper-localised advertising.
Fraud Detection	AI/ML pattern recognition in banking and healthcare; real-time risk scoring and transaction blocking.
Healthcare / Medicine	Patient profiling, real-time IoT wearable monitoring, disease-pattern identification for better treatments.

Key Challenges

- **Talent gap:** Acute shortage of skilled data professionals.
- **Data integration:** Synchronising disparate, heterogeneous data sources.
- **Storage complexity:** Managing scale, redundancy, and retrieval at volume.
- **Security and privacy:** Mitigating risks of breaches and compliance violations.

■ Quick-Reference Cheat Sheet

4 Vs: Volume · Velocity · Variety · Veracity | **4 Analytics:** Descriptive → Diagnostic → Predictive → Prescriptive | **5 Rs (Quality):** Relevancy · Recency · Range · Robustness · Reliability | **5 Layers:** Sources → Ingestion → Storage → Processing → Consumption | **Cloud Models:** IaaS · PaaS · SaaS | **NoSQL vs RDBMS:** Flexible schema, no JOINS, fault-tolerant replication